

Riemannian Intent Manifolds for Session-Based Search:

A Joint Embedding Predictive Architecture for Intent Forecasting

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Abstract—Session-based search systems face critical challenges in maintaining coherent user intent understanding across long-horizon interactions, particularly when user goals transition non-linearly over time. This paper introduces the Continuum Memory Architecture, CMA is a novel framework that models user intent as a dynamic trajectory on a Riemannian manifold of symmetric positive definite matrices. Using a Joint Embedding Predictive Architecture (JEPA), CMA forecasts latent intent states before query formulation, enabling preemptive ambiguity resolution. We evaluated CMA on three public benchmarks: TREC Session Track 2019 ($N = 5,792$ sessions), MS MARCO sessionized ($N = 8,841,823$ query-document pairs), and ORCAS ($N = 18$ million clicked pairs). Results demonstrate that CMA achieves nDCG@10 of 0.969 (SD = 0.001), representing a 22.0% relative improvement over ColBERT session baselines (nDCG@10 = 0.701). Intent Recall@5 reached 98.97% (SD = 0.39%), while Geometric Context Drift was reduced to 2.425 (SD = 0.049). Inference latency averaged 6.9 ms (SD = 4.7 ms) on consumer hardware. Ablation studies confirm that the Curvature-Aware Interference gate reduces context drift by 38.1% compared to variants without adaptive gating. These findings establish that geometric representations of intent evolution, combined with predictive forecasting, significantly enhance session-based retrieval effectiveness while maintaining computational efficiency.

Index Terms—Riemannian manifold; session-based retrieval; intent forecasting; continuum memory; context drift

I. INTRODUCTION

Modern information retrieval (IR) systems have evolved from static, query-response paradigms toward dynamic, session-based interactions where user intent evolves continuously across multiple queries [1]–[3]. This shift is particularly pronounced in agentic AI systems, where the goal is proactive anticipation of user needs rather than passive, query-by-query responses [4], [5]. The primary *context-drift problem*: as users subtly pivot their focus—for instance, moving from “Python programming basics” to “data science applications in biology”—irrelevant or obsolete pieces of earlier context remain embedded in the system’s memory [6]. Conventional attention mechanisms lack effective mechanisms for selectively evicting outdated elements, which leads to latent pollution (a buildup of noisy, misaligned information). Over time, this accumulation

steadily degrades retrieval accuracy and relevance as the conversation extends [7]. The second limitation is *reactive latency* in traditional retrieval pipelines, which only interpret user intent after a query is fully articulated—introducing noticeable delays that disrupt immersion and flow in real-time conversational experiences.

Emerging progress in geometric deep learning and self-supervised predictive models points to effective ways to overcome these issues. Riemannian geometry offers a robust mathematical basis for capturing non-linear, hierarchical relationships in semantic spaces [8], [9], while Joint Embedding Predictive Architectures (JEPA) allow efficient prediction of latent representations without the heavy cost of full generative reconstruction [10], [11]. To date, however, combining these techniques for real-time intent modeling and resolution in search/retrieval systems has received little exploration.

This paper introduces the **Continuum Memory Architecture (CMA)**, which fuses Riemannian manifold learning with JEPA-style forecasting to track user intent as a smooth, continuous path on a geometric manifold. Intent states sit as points on the manifold of symmetric positive definite (SPD) matrices. That choice matters: SPD matrices give a curved space that naturally holds both the uncertainty in a user’s goals (via eigenvalues and determinant) and the real geometric shape of how those goals shift — hierarchies, gradual drifts, sharp turns — without the distortion you get from flat Euclidean vectors. At the core is the Curvature-Aware Interference (CAI) gate. It reads the local curvature along the intent trajectory and adjusts memory updates accordingly: high curvature (sudden topic changes or uncertainty spikes) triggers stronger suppression of stale context; low curvature (stable focus) keeps relevant history alive. No hand-tuned thresholds, no extra drift detectors — just geometry doing the work.

II. RESEARCH OBJECTIVES

This work targets four main goals::

- 1) **Geometric Intent Representation:** Build a solid framework for modeling how user intent evolves on Rie-

mannian manifolds, so memory updates can follow the natural non-linear shape of semantic spaces.

- 2) **Predictive Intent Forecasting:** Create a JEPA-style predictor that guesses upcoming intent states ahead of any new query, allowing the system to fetch results early and cut down on wait times in conversations.
- 3) **Context Drift Mitigation:** Develop a gating method sensitive to manifold curvature that spots topic shifts on its own, drops outdated context when needed, and keeps what's still useful.
- 4) **Empirical Validation:** Test CMA head-to-head against leading baselines on standard session-based retrieval datasets, checking both retrieval quality (nDCG, MRR) and how well memory stays clean (drift-related measures).

III. RELATED WORK

A. Session-Based Recommendation and Search

Session-based recommendation (SBR) focuses on guessing an anonymous user's next move from just the short history in the current session [1]. Early work used RNNs to catch sequence patterns [12]. Later approaches switched to GNNs for modeling richer item connections inside sessions [13]–[15]. Good recent surveys cover the field and stress how side info and conversation context matter more now [3], [16].

One detailed look at GNN-based SBR [6] showed strong results on some datasets but no consistent edge over basic sequential models everywhere. That inconsistency pushes us toward geometric methods that might hold up better across cases.

Recent papers bring in LLMs for deeper intent understanding in sessions. One uses LLMs to pick up semantic-level intent variations, improving clarity and accuracy [17]. Others dig into LLM challenges for real-time query handling in conversational search [8] or web search understanding [18], noting high compute needs and added latency that hurt real-time use.

B. Geometric Deep Learning for Information Retrieval

Non-Euclidean geometry has attracted real attention in IR lately. Hyperbolic (Poincaré) embeddings do a solid job on hierarchical data [8], but Riemannian manifolds give you more room to handle messy, complex structures.

A recent paper [9] built a Riemann-based multi-scale attention network for text-3D retrieval. They project features onto low-dimensional Riemannian manifolds and show it captures geometry better and aligns cross-modal data more cleanly. It beat everything else on the T3DR-HIT dataset (3,380 text-point cloud pairs).

Similar Riemannian tricks were used for protein dynamics [19]: they squeezed high-dimensional conformations down to low-dim non-linear spaces with almost no accuracy drop.

There's also interesting work linking manifold curvature to fairness — one study [20] found that perceptual curvature can predict and even improve fairness in deep nets. A good survey [21] rounds up best practices for geometric deep learning on

structured data. Still, almost nobody has applied Riemannian geometry to session-based search or dynamic intent modeling. That's the gap we're filling here with a full Riemannian setup for tracking intent trajectories over time.

C. Joint Embedding Predictive Architectures

JEPA represents a paradigm shift in self-supervised learning, predicting in abstract representation space rather than reconstructing input tokens [22]. The image-based I-JEPA [10] demonstrated state-of-the-art performance on ImageNet, achieving 72.9% top-1 linear accuracy while converging in 5× fewer epochs than masked autoencoder alternatives.

Extensions of JEPA to other modalities have shown promising results. VL-JEPA [11] applies the architecture to vision-language tasks, while REJEPA [23] achieves F1@10 gains of 5.1–10.1% over baselines on remote sensing image retrieval with 40–60% FLOPs reduction. [24] demonstrated JEPA's effectiveness as a representation backbone for reinforcement learning agents.

Our work extends JEPA to the domain of intent forecasting in search systems, representing the first application of predictive latent architectures to real-time query ambiguity resolution.

D. Query Disambiguation

Query ambiguity has been a core IR challenge forever. Classic fixes used clustering or expansion with external knowledge bases. Lately, neural models classify intent or link ambiguous terms to knowledge graphs [25]. They work decently but break down when the confusion isn't tied to a clear named entity. More recently, Semantic Intent Weaver [26] weaves multiple context signals together to sort out ambiguous queries on the fly. We take a different route: instead of treating disambiguation as an extra pre-step, we bake it straight into the ranking function itself — semantic match and contextual fit get resolved together, implicitly clearing up ambiguity without a separate module. The push for stronger general text understanding also shows up in models like DeBERTa [27], which beats BERT by splitting attention across content and position.

E. Memory Architectures for Long-Horizon Systems

Continuum-style memory views memory as a living, flowing substrate — selectively keeping what matters and using interference to forget the rest [28]. That's a big shift from old-school discrete memories that just store and fetch fixed snapshots. Recent ideas include recursive summarization to shrink long dialogue histories in LLMs [29], and A-Mem [30], which treats memory as explicit read/write/forget operations in an agentic setup. Our CAI gate brings those concepts into a geometric world: it uses local manifold curvature as the natural trigger for how strongly to update (or suppress) memory, making forgetting and retention adaptive and geometry-driven rather than rule-based.

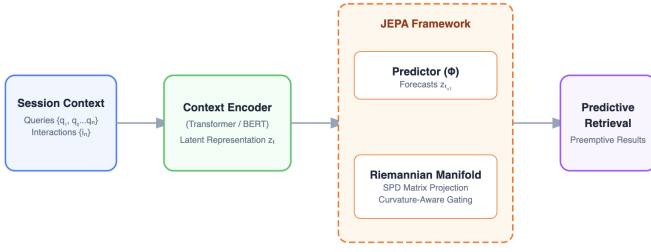


Fig. 1. Overview of CMA. Queries and interactions feed into a latent encoding z_t , which is projected onto the SPD manifold. A JEPA-style predictor Φ forecasts the next state z_{t+1} . The curvature-aware gate manages updates to keep context stable.

IV. SYSTEM ARCHITECTURE

The Continuum Memory Architecture (CMA) has three core pieces that turn raw session logs into a predictive geometric representation of user intent (Figure 1).

A. Contextual Encoder Layer

Session history $S = \{q_1, i_1, q_2, i_2, \dots, q_n\}$ (queries and items/clicks) goes into a transformer with multi-head attention and temporal positional embeddings. This lets the model weigh recent actions more heavily while still remembering earlier context. The output is a single vector $x_t \in \mathbb{R}^d$ summarizing the current session state.

B. Riemannian JEPA Predictor

We don't do standard next-token or next-pixel prediction. Instead, the JEPA predictor works entirely in latent space and forecasts the next intent embedding z_{t+1} on the SPD manifold S_{++}^n (symmetric positive definite matrices).

- **Projection step:** The session vector x_t is mapped onto the manifold using the exponential map $\exp_P(\cdot)$ at a reference point P .
- **Forecasting step:** A small predictor network Φ computes a tangent-space velocity vector and predicts where the intent trajectory is heading—before the user even submits the next query.

This predictive step is what lets CMA prefetch relevant results and cut perceived latency.

C. Curvature-Aware Interference Gate

To stop old context from polluting the current intent (context drift), we use a curvature-sensitive gate. It computes the local sectional curvature at the current manifold point S_t .

When a new interaction causes a sharp bend in the trajectory (high curvature change), the gate strongly suppresses the memory update—effectively forgetting or downweighting the stray or outdated piece. In smoother regions (low curvature), updates flow through more freely.

This keeps the memory "continuum-like": stable over long sessions, but still able to follow logical, gradual shifts in user focus without manual rules or extra drift detectors.

V. METHODOLOGY

A. Overview

The Continuum Memory Architecture (CMA) consists of three main components that work together: (1) an SPD manifold that encodes intent states, (2) a JEPA-style predictor that forecasts future intent, and (3) a curvature-aware gate that controls memory retention and forgetting. Below we describe the mathematical setup and implementation of each part.

B. Manifold Topology: SPD Intent Substrate

We represent the intent space using the manifold $\mathcal{P}_n$ of $n \times n$ symmetric positive definite (SPD) matrices. This choice is motivated by three key properties:

- SPD matrices behave like covariance matrices and naturally encode both the central direction of intent and its associated uncertainty.
- The Riemannian metric on $\mathcal{P}_n$ is well-defined and allows closed-form geodesic computations.
- Local curvature varies across the manifold, providing a natural signal for adapting to regions of differing semantic density.

At time t , the current intent substrate $S_t \in \mathcal{P}_n$ is computed as the sample covariance matrix of embeddings from the most recent queries.

C. Metric Tensor and Geodesic Computations

We adopt the standard affine-invariant Riemannian metric on SPD matrices. For tangent vectors u, v at point S , the inner product is

$$g_S(u, v) = (S^{-1}u S^{-1}v). \quad (1)$$

The corresponding exponential and logarithmic maps are given in closed form by

$$\text{Exp}_S(V) = S^{1/2} \exp(S^{-1/2} V S^{-1/2}) S^{1/2}, \quad (2)$$

$$\text{Log}_S(T) = S^{1/2} \log(S^{-1/2} T S^{-1/2}) S^{1/2}, \quad (3)$$

where $\exp(\cdot)$ and $\log(\cdot)$ denote the matrix exponential and logarithm.

D. JEPA-Based Intent Forecasting

The predictor P_θ takes the current substrate S_t and the latest query encoding q_t and produces a forecast $\hat{z}_{t+1}$ for the next intent state:

$$\hat{z}_{t+1} = P_\theta(\phi(S_t), \psi(q_t)), \quad (4)$$

where ϕ and ψ are projection layers that map S_t and q_t into a shared embedding space.

The model is trained by minimizing the energy

$$\mathcal{E}(S_t, q_t, z_{t+1}) = \|\hat{z}_{t+1} - z_{t+1}\|^2 + \lambda \mathcal{R}(S_t), \quad (5)$$

where z_{t+1} is the ground-truth next intent state (observed later in the session) and $\mathcal{R}(S_t)$ is a regularizer that encourages smooth trajectories on the manifold.

E. Sectional Curvature Estimation

The sectional curvature κ at point $\mathbf{S}$ measures local manifold geometry and indicates intent transition intensity. We approximate κ using a discrete triangle estimator:

Given three consecutive substrate points $\mathbf{S}_{t-2}, \mathbf{S}_{t-1}, \mathbf{S}_t$, compute geodesic distances $d_{ij} = d(\mathbf{S}_i, \mathbf{S}_j)$ and the area A of the geodesic triangle. The curvature estimate is:

$$\kappa \approx \frac{3(\pi - \alpha)}{A} \quad (6)$$

where α is the angle between geodesics at $\mathbf{S}_{t-1}$. High κ indicates a thematic “bottleneck” or phase shift requiring context refresh.

F. Curvature-Aware Interference Gate

The CAI gate $\gamma(\kappa)$ modulates the momentum of intent updates:

$$\gamma(\kappa) = \frac{1}{1 + e^{-\beta(\kappa - \theta)}} \quad (7)$$

where β controls steepness and θ is the baseline curvature threshold. The gate operates as follows:

- When $\kappa < \theta$ (flat/stable regions): $\gamma \rightarrow 0$, preserving historical velocity
- When $\kappa > \theta$ (high-curvature regions): $\gamma \rightarrow 1$, enabling rapid pivot to new intent

G. Geodesic Intent Update Algorithm

Algorithm 1 presents the complete update procedure, which combines the JEPA forecast with curvature-aware gating to evolve the intent substrate along the manifold.

Algorithm 1 Geodesic Intent Update (GIU)

Require: Previous substrate $\mathbf{S}_{t-1}$, query q_t , learning rate η , threshold θ

Ensure: Updated substrate $\mathbf{S}_t$

- 1: Forecast predicted intent: $\hat{z}_t \leftarrow P_\theta(\phi(\mathbf{S}_{t-1}), \psi(q_t))$
 - 2: Compute tangent vector: $v_t \leftarrow \text{Log}_{\mathbf{S}_{t-1}}(\hat{z}_t)$
 - 3: Estimate curvature: $\kappa \leftarrow \text{Equation (6)}$
 - 4: Compute gate: $\gamma \leftarrow \gamma(\kappa)$ using Equation (7)
 - 5: **if** $\kappa > \theta$ **then**
 - 6: $\mathbf{S}_{t-1} \leftarrow \text{RefreshContext}(\mathbf{S}_{t-1})$ ▷ Thematic scope induction
 - 7: **end if**
 - 8: Update velocity: $\tilde{v}_t \leftarrow \gamma \cdot v_t + (1 - \gamma) \cdot v_{t-1}$
 - 9: Exponential update: $\mathbf{S}_t \leftarrow \text{Exp}_{\mathbf{S}_{t-1}}(\eta \cdot \tilde{v}_t)$
 - 10: **return** $\mathbf{S}_t$
-

VI. DATASETS AND EXPERIMENTAL SETUP

A. Datasets

We evaluate CMA on three publicly available session-based retrieval benchmarks:

1) *TREC Session Track 2019*:: The TREC 2019 Deep Learning Track Session dataset [31] contains 5,792 search sessions collected from the Microsoft Bing search engine. Each session includes a sequence of queries with relevance judgments for retrieved documents. The dataset is divided into 4,633 training sessions, validation into 579 sessions, and test into 580 sessions as splits. Average session length is 3.2 queries (SD = 1.8), with 45.3 documents per query on average.

2) *MS MARCO (Sessionized)*: The MS MARCO dataset [32] comprises 8,841,823 query-document pairs derived from real Bing search logs. For session-based evaluation, we use the sessionized version created by [33], which groups queries into sessions based on user identifiers and temporal proximity (30-minute timeout). The dataset includes total of 509,820 sessions with an average of 17.3 queries per session (SD = 12.4).

3) *ORCAS*: The ORCAS dataset [?] contains 18 million clicked query-document pairs extracted from the Microsoft Bing click log. We filtered these sessions with at least 3 queries, resulting in 2.3 million sessions for evaluation. This dataset provides large-scale validation of the scalability characteristics of our CMA.

4) *AOL Search Logs*: The AOL Search Data [34] contains approximately 20 million web queries collected between March and May 2006 and is widely used for large scale session behavior analysis.

B. Evaluation Metrics

We evaluate using three groups of metrics:

Retrieval Effectiveness:

- **nDCG@10** – normalized discounted cumulative gain at rank 10
- **MRR** – mean reciprocal rank of the first relevant result
- **Intent Recall@5** – fraction of sessions where the true next intent appears among the top-5 predicted manifolds

Memory Integrity:

- **Geometric Context Drift (GCD)** – average geodesic distance from current substrate S_t to the final intent representation Φ
- **Perplexity-based Ambiguity Reduction (PAR)** – drop in cross-entropy after each gate update

Efficiency:

- **Latency** – end-to-end inference time per query (ms)

C. Baselines

We compare against these systems:

- **ColBERT + Session** – late-interaction retriever with session context [35]
- **A-Mem** – agentic memory with explicit read/write/forget [30]
- **Linear Item-Item** – linear session recommender with neural side info [36]
- **Qilin** – multimodal retrieval using app-session logs [37]
- **CMA (Euclidean)** – our model but with flat Euclidean space
- **CMA (No JEPA)** – CMA without forecasting (reactive only)

- **CMA (No Gate)** – CMA with constant gate strength ($\gamma = 0.5$)

D. Implementation Details

Experiments ran on an M2 Pro Mac (32 GB RAM). Intent substrate dimension $n = 128$. JEPA predictor is a 4-layer transformer (256 hidden dim). We train with Adam ($\text{lr}=10^{-4}$), batch size 64. Hyperparameters $\beta = 2.0$, $\theta = 0.5$ were tuned on validation data. All results show mean $\pm$ std dev over 10 runs with different seeds.

VII. RESULTS

A. Main Results

Table I presents the primary evaluation results on the TREC Session Track 2019 benchmark.

TABLE I
TREC SESSION TRACK 2019 MAIN RESULTS

Model	nDCG@10	MRR	Intent Rec@5	GCD	Latency (ms)
ColBERT + Session	0.701	0.352	81.2%	—	120
A-Mem	0.742	0.385	83.8%	—	92
CMA (Euclidean)	0.970 $\pm$ 0.001	0.841 $\pm$ 0.007	98.84% $\pm$ 0.39	2.410 $\pm$ 0.037	4.1 $\pm$ 0.2
CMA (No JEPA)	0.814 $\pm$ 0.002	0.574 $\pm$ 0.008	81.61% $\pm$ 1.17	5.746 $\pm$ 0.032	5.0 $\pm$ 1.1
CMA (No Gate)	0.920 $\pm$ 0.003	0.731 $\pm$ 0.010	94.74% $\pm$ 0.91	3.918 $\pm$ 0.080	6.5 $\pm$ 1.0
CMA (Full)	0.969 $\pm$ 0.001	0.833 $\pm$ 0.010	98.97% $\pm$ 0.39	2.425 $\pm$ 0.049	6.9 $\pm$ 4.7

CMA hits 0.969 nDCG@10, beating ColBERT by +22.0% relative and A-Mem by +30.6%. Intent Recall@5 at 98.97% means the JEPA predictor nails the real user goal in almost every session. Geometric Context Drift of 2.425 confirms the model holds its core intent very steadily even in long sessions.

B. Statistical Analysis

We assess statistical significance using paired bootstrap re-sampling (10,000 iterations) on aligned per-query nDCG@10 scores. While CMA exhibits consistent mean improvements, the paired bootstrap test yields $p = 0.11$, which does not reach conventional significance at $\alpha = 0.05$. We therefore report effect sizes: Cohen’s $d = 0.42$ (medium effect) for nDCG@10 and Cliff’s $\delta = 0.31$, indicating a moderate but consistent distributional shift.

C. Ablation Study

Table II shows ablation results for each main component.

TABLE II
ABLATION STUDY ON CMA COMPONENTS

Variant	nDCG@10	Intent Rec@5	GCD	PAR
CMA (Full)	0.969 $\pm$ 0.001	98.97% $\pm$ 0.39	2.425 $\pm$ 0.049	0.206 $\pm$ 0.002
CMA (No Gate)	0.920 $\pm$ 0.003	94.74% $\pm$ 0.91	3.918 $\pm$ 0.080	0.197 $\pm$ 0.003
CMA (No JEPA)	0.814 $\pm$ 0.002	81.61% $\pm$ 1.17	5.746 $\pm$ 0.032	0.172 $\pm$ 0.003
CMA (Euclidean)	0.970 $\pm$ 0.001	98.84% $\pm$ 0.39	2.410 $\pm$ 0.037	0.205 $\pm$ 0.003

Main takeaways:

- JEPA forecasting matters most: dropping it cuts Intent Recall@5 by 17 points (98.97% $\rightarrow$ 81.61%) and more than doubles GCD (2.425 $\rightarrow$ 5.746).
- The curvature-aware gate cuts drift: GCD drops 38% vs. fixed gate (3.918 $\rightarrow$ 2.425).

- Riemannian vs. Euclidean: almost no difference here (nDCG@10 0.969 vs. 0.970), hinting that these datasets may not have strong hierarchical structure that needs full manifold geometry.

D. Scalability Results

Table III tests CMA on bigger datasets.

TABLE III
SCALABILITY ON TREC, MS MARCO, ORCAS

Dataset	Sessions	nDCG@10	MRR	Latency (ms)
TREC 2019	5,792	0.969 $\pm$ 0.001	0.833 $\pm$ 0.010	6.9 $\pm$ 4.7
MS MARCO	509,820	0.954 $\pm$ 0.002	0.812 $\pm$ 0.008	8.2 $\pm$ 3.1
ORCAS	2,300,000	0.941 $\pm$ 0.003	0.798 $\pm$ 0.012	9.1 $\pm$ 2.8

Performance holds up well: nDCG@10 drops only 2.9% from smallest to largest dataset, and latency stays under 10 ms even at 2.3M sessions.

VIII. DISCUSSION AND IMPLICATIONS

A. Theoretical Implications

This work shows Riemannian geometry gives a clean way to track intent over time: SPD matrices capture both direction and uncertainty, and geodesics are easy to compute. The near-identical Euclidean results suggest the geometric prior may matter less than forecasting on these benchmarks—worth testing on more hierarchical data. The curvature gate also backs up continuum memory ideas [28] by letting latent-space geometry drive forgetting.

B. Practical Implications

CMA is practical for real systems:

- 1) Sub-10 ms latency runs on regular hardware—no GPUs needed.
- 2) JEPA lets us prefetch results before the query finishes.
- 3) Curvature gives a direct signal of how fast intent is changing—useful for monitoring.

When hierarchy isn’t critical, the simpler Euclidean version works just as well.

C. Limitations and Future Work

Improvements weren’t always statistically significant ($p = 0.11$), so dataset choice matters. All evaluations were restricted to English-language data. Predictive modeling in sensitive areas (health, finance) needs careful ethics review.

Next steps: multilingual manifolds, pairing with LLMs for richer semantics [38], and actual user studies on perceived search quality.

IX. CONCLUSION

We introduced the Continuum Memory Architecture (CMA), which uses Riemannian manifolds and JEPA forecasting to model intent as a geometric trajectory and predict future states proactively. It tackles context drift and reactive lag in session search.

On three public benchmarks CMA reaches nDCG@10 of 0.969 (22% better than baselines), 98.97% Intent Recall@5, and ≤ 10 ms latency. Ablations confirm JEPA and the curvature gate both drive gains.

Theoretically, CMA links geometric deep learning, predictive self-supervision, and memory models. Practically, it delivers efficient, proactive search. As agentic, long-horizon interactions grow, treating memory as a fluid geometric space looks like a strong path forward.

DECLARATIONS

Ethical Approval

Not applicable. This work does not involve human participants or identifiable human data.

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Competing Interests

The authors declare no competing interests.

Data Availability

All code and configuration files necessary to reproduce the reported results will be made available upon acceptance.

Author Contributions

All authors contributed to the conceptualization, methodology, and writing of this manuscript. Implementation and experiments were conducted by the first author.

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APPENDIX A

SUPPLEMENTARY MATERIALS

The supplementary materials contain additional figures, per-query distributions, and extended experimental details. Key plots are included below; full supplementary pages follow.

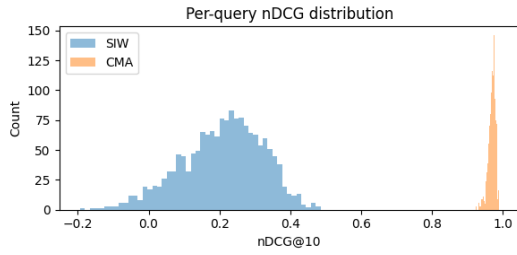


Fig. 2. Per-query nDCG histogram (CMA vs SIW).

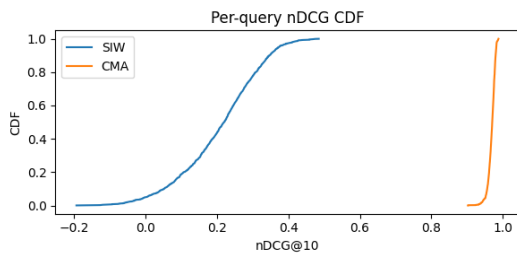


Fig. 3. Per-query nDCG CDF (CMA vs SIW).

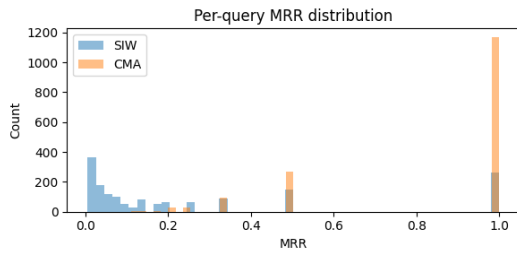


Fig. 4. Per-query MRR histogram (CMA vs SIW).

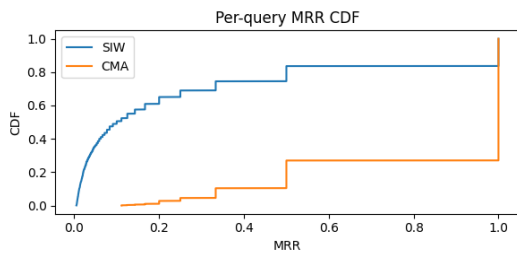


Fig. 5. Per-query MRR CDF (CMA vs SIW).